

ZHEYUN WAN

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EDUCATION

Nanchang University (GPA: 3.49/4.0) Supervisor: Associate Professor. Jingbo Wei Master in Computer Science	Nanchang, China Sep 2020 – July 2023
Jiangxi University of Finance and Economics (GPA: 2.6/5.0) B.S. in Software Engineering	Nanchang, China Sep 2015 – July 2019

PUBLICATIONS

* indicates equal contribution

† indicates corresponding author.

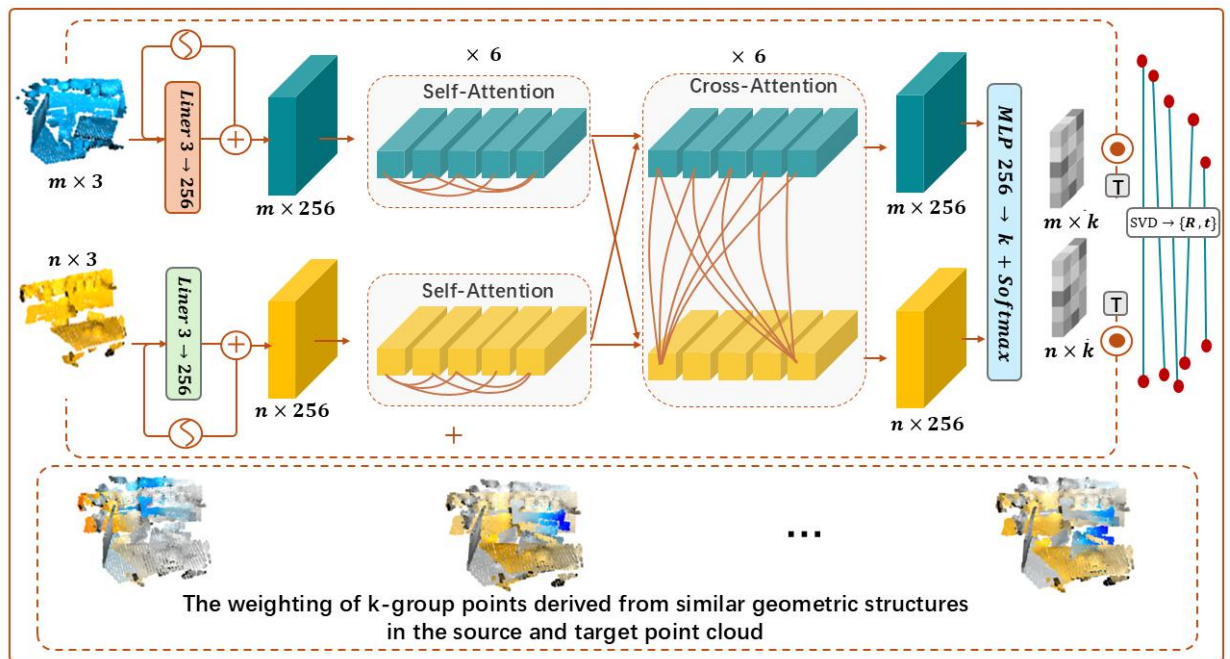
[PLOS ONE]	A Strengthened Soft-Point Registration Network for LiDAR Point Cloud Registration Yijie Chen, Bin Tian, Zeyun Wan †, Shichao Fan3, Mingxing Chen, Yi Wu, Chunquan Li PLOS ONE, IF: 2.8, 2026
[Neurocomputing]	ShapeGPT and PointPainter for fast zero shot text-to-3d point cloud generation Zeyun Wan *, Shichao Fan*, Huan Zhao, Jiawei Liu, Ying Chen Neurocomputing, IF: 5.5, 2025
[Remote Sensing]	Learning Contours for Point Cloud Completion Jiabo Xu, Zeyun Wan , Jingbo Wei MDPI Remote Sensing, IF: 5.5, 2023
[IEEE TGRS]	GLORN: Strong Generalization Fully Convolutional Network for Low-Overlap Point Cloud Registration Jiabo Xu, Yukun Huang, ZeYun Wan , Jingbo Wei IEEE Transactions on Geoscience and Remote Sensing, IF: 7.5, 2022

RESEARCH INTERESTS

Generative AI, 3D Reconstruction, Point-Cloud Processing

RESEARCH EXPERIENCE

- School of Information Engineering, Nanchang University* *Nanchang, China*
A soft-point registration network for LiDAR point cloud registration July 2024 – June 2026
Co-developed SPRN, an end-to-end, feature-matching-free and RANSAC-free framework for partial LiDAR point-cloud registration. The network uses six self-attention layers to encode intra-cloud geometry and six cross-attention layers to exchange source–target information, generates (k=8) paired soft points as attention-weighted centroids, and estimates the rigid transformation through differentiable Kabsch–Umeyama/SVD.
 - Contributed to the soft-point correspondence module, replacing explicit or implicit pairwise feature matching and RANSAC-based outlier rejection with learned soft-point aggregation and direct pose estimation under a transformation loss.
 - Evaluated SPRN on 3DMatch, 3DLoMatch, KITTI, and WHU-TLS, achieving 92.8% registration recall on 3DMatch and 75.0% on 3DLoMatch. On 3DLoMatch, SPRN matched GeoTransformer’s recall while reducing RRE from 2.641° to 2.278° and RTE from 0.081 m to 0.069 m, with a reported runtime of 0.069 seconds per pair on an RTX 3090.



- School of Computer and Information Engineering, Nanchang Institute of Technology* *Nanchang, China*

Fast zero shot text-to-3d point cloud generation using GPT

March 2024 – Aug 2024

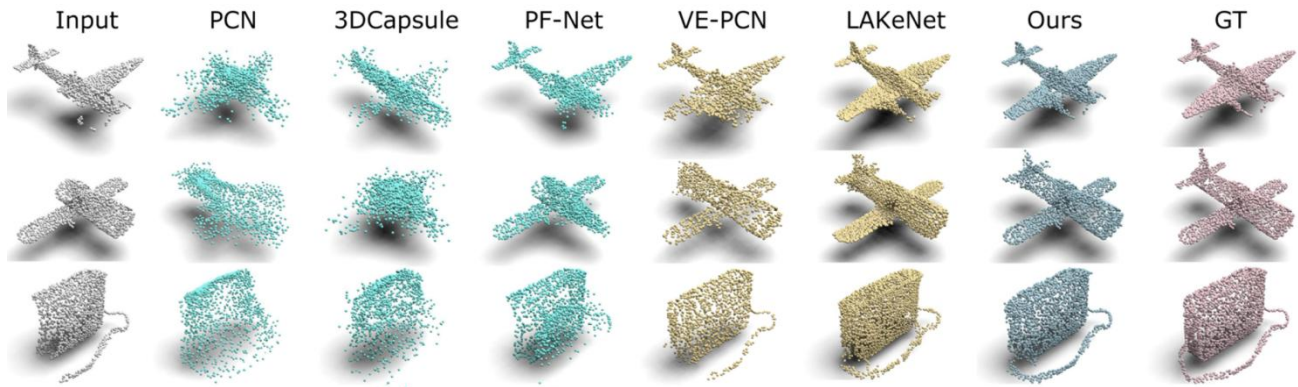
Developed the two-stage ShapeGPT text-to-3D pipeline: Stage 1 uses a GPT-style transformer to generate coarse voxel priors directly from raw text prompts; Stage 2 employs a lightweight PointPainter refinement network to convert those voxels into detailed point clouds

- Designed and implemented the ShapeGPT architecture to encode text prompts into 3D shape priors, reducing inference time compared to prior methods.
- Implemented the PointPainter module to “paint” geometric detail onto the ShapeGPT voxels, boosting reconstruction fidelity and achieving state-of-the-art improvements on standard benchmarks (e.g. ShapeNet), with up to a 20% reduction in error versus previous zero-shot methods.



- School of Mathematics and Computer Sciences, Nanchang University* *Nanchang, China*

1. Designed a novel end-to-end “point $\rightarrow$ voxels $\rightarrow$ points” completion network by integrating transformer-based encoder and decoder modules, improving model compactness over VE-PCN by eliminating extra multi-scale branches while preserving rich structural priors for completion.
2. Incorporated an implicit voxel-validity classifier with a contour-learning loss to capture salient edges and irregular surfaces, yielding more accurate detail reconstruction on ShapeNet and KITTI benchmarks compared to standard transformer-only decoders.



LANGUAGE SKILLS

- English: IELTS 7.5